Monitoring Rice Growth Status and Grain Yield by Combining With Various Growth Parameters Using UAV-Based Hyperspectral Images

Zhonglin Wang, Chuanhai Shu, Huilai Yin, Yufei Xie, Tao Liu, Feng Yang, Feijie Li, Yongjian Sun, Zhiyuan Yang, and Jun Ma

***Abstract*—Timely and accurate monitoring of the comprehensive growth status in rice is crucial for the precise estimation of grain yield and the guarantee of food security. The comprehensive growth monitoring indicator (CGMI) is constructed using various growth parameters to assess crop growth levels. However, the multiparameter-driven CGMI is prone to fluctuation and uncertainty in high-dimensional data, limiting its stability and the physiological interpretability of grain yield. This study proposed a novel method, the information fusion stability (IFS) method, to construct CGMI, explore its accuracy and stability for monitoring rice growth status, and investigate its physiological response to grain yield. The entropy weight method (EWM), the coefficient of variation (CV) method, and the IFS method were used to calculate the $CGMI_{EWM}$, $CGMI_{CV}$, and $CGMI_{IFS}$. The permutation importance algorithm was used to select more important vegetation indices (VIs) to develop estimation models of three CGMIs. A yield-decline-ratio-based thresholding strategy was adopted to assess the comprehensive growth level, and the optimal CGMI estimation model was incorporated to draw the spatial distribution map of CGMI and grain yield. The CGMIs were estimated using VIs selected by feature selection, and the random forest (RF) regression models of $CGMI_{IFS}$ had the best test accuracy with root-mean-square errors (RMSEs) of 0.030, 0.045, and 0.044 at the tillering, jointing, and full-heading stages, respectively. A quadratic relationship between the three CGMIs and grain yield was determined for three growth stages. A yield-decline-ratio-based thresholding strategy was used to classify the comprehensive growth status into three levels: weak, suitable, and excessive. The comprehensive growth level can respond sensitively to differences in nitrogen rates. Grain yield estimated using hyperspectral images at the full-heading stage was closest to the real grain yield (5.7–10.2 $t \cdot hm^{-2}$). As the growth stage progressed, the CGMI threshold increased, which explained the nitrogen transport, photosynthetic matter accumulation, and organ formation required for high yields in rice. This study confirmed the stability of the IFS method in constructing CGMI and the reliability of assessing rice growth levels, which provides a valuable solution for nitrogen management and grain yield estimation.**

***Index Terms*—Comprehensive growth monitoring indicator (CGMI), grain yield, hyperspectral image, information fusion stability (IFS) method, rice, uncrewed aerial vehicles (UAVs).**

I. Introduction

RICE (*Oryza sativa* L.) is regarded as one of the most important food crops in global agriculture and is consumed by over half of the world's population [1]. With an increasing global population and escalating resource constraints, the sustainable development of rice yield has been considered critical for ensuring food security and maintaining social stability. The timely, accurate, and nondestructive monitoring of rice growth status using uncrewed aerial vehicles (UAVs) remote sensing has been proven to be an effective approach to evaluate crop productivity and quality [2], [3], [4], [5], [6], [7]. This approach could provide technical support for optimizing nitrogen fertilizer management, improving grain yield and quality, and informing market-oriented trade strategies.

Numerous studies have quantitatively assessed crop growth status by monitoring growth parameters using remote sensing data, including morphological traits [8], [9], [10], physiological and biochemical indicators [11], [12], [13], and 3-D structures [14], [15]. Morphological traits are understood to include crop height (CH), leaf area index, aboveground biomass (AGB), and canopy cover. Physiological and biochemical parameters primarily refer to photosynthetic pigments, water content, and nitrogen content of crops. Both morphological traits and physiological–biochemical parameters can be accurately estimated using vegetation indices (VIs), color indices (CIs), and texture indices [16], [17], [18], [19]. With advancements in phenomics and deep learning, 3-D structures have been reconstructed from RGB images and LiDAR data, enabling more intuitive quantification of growth status [20], [21]. However, single growth parameters represent only specific and limited aspects of crop characteristics, making it difficult to comprehensively evaluate crop growth status and thereby constraining the growth level assessment based on remote sensing data. When multiple growth parameters are

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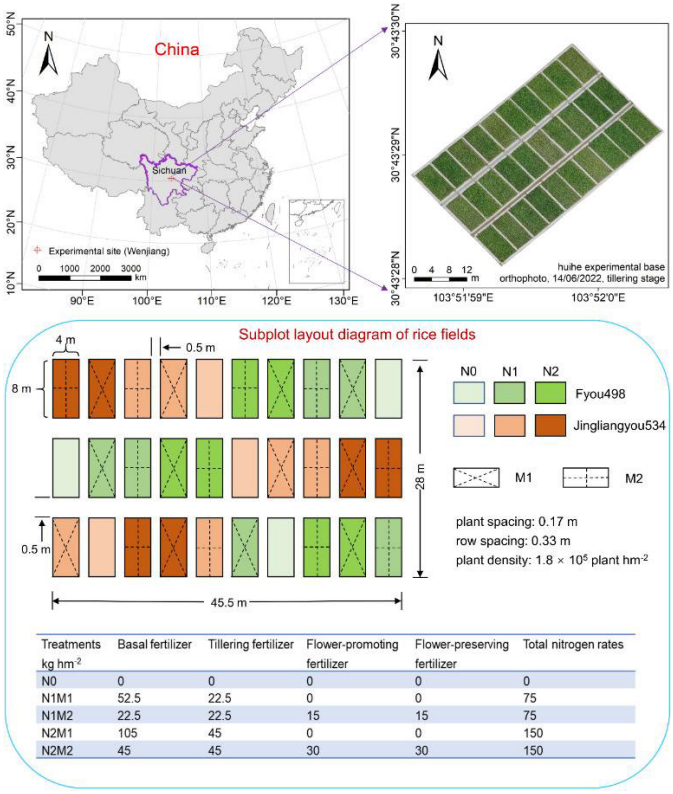
simultaneously retrieved from remote sensing data, it remains challenging to reflect the comprehensive growth levels, as the inversion of different parameters requires substantial computational resources and time, along with specific evaluation criteria [22], [23].

Previous studies have proposed comprehensive growth monitoring indicators (CGMIs) by combining morphological traits, physiological and biochemical parameters, and canopy structural information to evaluate comprehensive crop growth status [22], [24], [25]. These studies have reported that VIs and CIs, derived from UAV- and satellite-based remote sensing, serve as key features for accurately estimating crop CGMIs. Importantly, the calculation methods of CGMIs and growth level classification strategies have been identified as a critical factor for growth assessment performance. Currently, CGMI calculation methods include the equal weighting method, the coefficient of variation (CV) method, the entropy weight method (EWM), and the fuzzy comprehensive evaluation method. Several studies have introduced novel computational methods and compared them with conventional approaches to enhance the estimation accuracy and evaluation performance of CGMIs. For example, Wang et al. [22] proposed a game theory combination objective weight method to comprehensively assess cotton growth levels. Yang et al. [26] have improved the estimation accuracy of maize CGMIs by integrating the technique for order preference by similarity to ideal solution with the CV method. The incorporation of more growth parameters into the CGMIs computation has been shown to improve the comprehensiveness and accuracy of crop growth assessments [27]. However, under the challenges posed by high-dimensional variability and uncertainty in multiparameter-driven growth assessments, constructing an unsupervised CGMI that balances information contribution and stability remains a critical issue in smart agriculture. To address these challenges, a comprehensive evaluation method is proposed in this study, which simultaneously considers indicator stability and information contribution while incorporating an adaptive weight mechanism. The main objectives of this study are: 1) to propose an information fusion stability (IFS) method to calculate the new CGMI by combining CH, AGB, plant water content (PWC), leaf chlorophyll content (CHL), leaf carotenoid content (CAR), leaf nitrogen content (LNC), and leaf carbon content (LCC); 2) to compare the variation in CGMIs derived from the CV, EWM, and IFS methods; 3) to develop the remote sensing monitoring models of CGMIs using machine learning regression algorithms; 4) to assess comprehensive growth levels using yield-decline-ratio-based thresholding strategy; and 5) to explore the physiological response of grain yield to CGMIs. The findings of this study are anticipated to promote quantitative assessment of rice growth in the field and differentiated nitrogen fertilizer management, thereby optimizing the accurate estimation of grain yield.

II. Materials and Methods

A. Experimental Design

Two field experiments were conducted in 2022 and 2023 at the Huihe Experimental Base of Sichuan Agricultural University, Chengdu, Sichuan, China (see Fig. 1). A randomized complete block design with three factors was adopted, consisting of 10 treatments with three replications, totaling 30 subplots. Two rice cultivars, Fyou498 and Jingliangyou534, were sown on April 7, 2022 and April 8, 2023, manually transplanted on May 13, 2022 and May 8, 2023, and harvested on August 31, 2022 and September 5, 2023. Three nitrogen levels (0, 75, and 150 kg · hm^{-2}) were applied in the form of urea (46.67% N). Two nitrogen application percentages were set: 1) 70% for the pretransplanting and 30% for the tillering stage (M1 = 7:3) and 2) 30% for the pretransplanting, 30% for the tillering stage, 20% for the jointing stage, and 20% for the booting stage (M2 = 3:3:2:2). Phosphorus fertilizer (75 kg · hm^{-2} as P_2O_5) and potassium fertilizer (150 kg · hm^{-2} as K_2O) were applied entirely at the pretransplanting stage. The planting configuration was fixed with a row spacing of 0.33 m and plant spacing of 0.17 m, corresponding to a planting density of 1.8×10^5 plants · hm^{-2}. Each subplot was 8 m in length and 4 m in width. Each subplot was separated by a soil ridge (approximately 0.4 m in width and 0.25 m in height), wrapped with plastic film to ensure independent water and fertilizer management.



Treatments	Basal fertilizer (kg hm^{-2})	Tillering fertilizer	Flower-promoting fertilizer	Flower-preserving fertilizer	Total nitrogen rates
N0	0	0	0	0	0
N1M1	52.5	22.5	0	0	75
N1M2	22.5	22.5	15	15	75
N2M1	105	45	0	0	150
N2M2	45	45	30	30	150

Fig. 1. Experimental location and subplot layout diagram of the rice field.

B. Data Acquisition

UAV-based hyperspectral images and growth parameters were acquired on the same day at the tillering, jointing, and full-heading stages to ensure temporal synchronization between spectral observations and ground-truth data. The tillering, jointing, and full-heading stages were selected

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TABLE I
COLLECTION DATES FOR HYPERSPECTRAL IMAGES, GROWTH PARAMETERS, AND GRAIN YIELD

Datasets	Growth stages	Collection dates	
		2022	2023
Hyperspectral images	Tillering stage	14/06 (32)	18/06 (41)
Gowth parameters	Jointing stage	06/07 (54)	08/07 (61)
	Full-heading stage	28/07 (76)	04/08 (88)
Grain yield	Maturity stage	31/08 (110)	05/09 (120)

Note: The numbers in parentheses indicate the number of days after rice transplanting.

because they represent critical periods of rice growth and development, which are closely associated with biomass accumulation, nutritional status, and grain yield formation. Immediately after hyperspectral image acquisition, three healthy and uniform rice plants were randomly sampled from each subplot, transported to the laboratory in water buckets, and processed by removing roots, separating leaves, stems, and panicles, and cleaning surface moisture, dust, and soil. Grain yield was collected at the maturity stage, with sampling dates listed in Table I. The detailed growth parameters and measurement methods are as follows.

1) Acquisition of Hyperspectral Images: Hyperspectral images were collected using a UAV (DJI Matrice 600 Pro, DJI Technology Company Ltd., Shenzhen, China) equipped with a push-broom hyperspectral camera (Resonon Pika L, Resonon Inc., Bozeman, MT, USA). The specific specifications of the hyperspectral camera and aircraft are provided in Table II. The camera covered the 400–1000 nm spectral range with a spectral resolution of 3.3 nm and a band interval of 4 nm. The lens had a field of view of 16.7° and an instantaneous field of view of 0.71 mrad. Flight parameters were set using DJI Pilot software (v2.5.1.15, DJI Technology Company Ltd., Shenzhen, China). Meanwhile, a high-precision RTK-assisted inertial navigation system was mounted on the aircraft to ensure flight stability and position accuracy, and to collect positioning and orientation system (POS) data. The aircraft was flown at 50 m altitude and 1.5 m $\cdot$ s^{-1} speed from north to south, with 80% forward overlap and 50% side overlap, resulting in a ground swath of 15.5 m. Flight missions were performed between 11:00 and 13:00 under sunny and cloudless weather. A 1 $\times$ 1 m calibration tarp with 30% reflectance was placed in the field for postprocessing reflectance correction.

2) Measurement of Growth Parameters: CH: CH was measured from the plant base to the highest leaf or panicle tip using a measuring tape.

AGB and PWC: Fresh weight of stems, leaves, and panicles was measured using an electronic balance. All samples were oven-dried at 105 °C for 0.5 h and dried at 70 °C until constant weight. The dry weight of samples was then weighed to calculate AGB and PWC using the following equation:

$$\mathrm{AGB}\left(\mathrm{kg\ m^{-2}}\right) = \frac{\text{plant dry weight}}{\text{per unit ground area}} \tag{1}$$

$$\mathrm{PWC}\,(\%) = \frac{\text{plant fresh weight-plant dry weight}}{\text{plant fresh weight}} \tag{2}$$

where plant dry weight is the sum of the dry weights of the stems, leaves, and panicles, and plant fresh weight is the sum of the fresh weights of the stems, leaves, and panicles.

Photosynthetic Pigments: Three fully expanded leaves were collected and shredded from the top of the rice plant. About 0.1 g of filamentous fresh leaves was put in the extraction solution of acetone: absolute ethanol: distilled water = 4.5:4.5:1 (volume ratio). The samples were placed in a dark environment at room temperature until all the green leaves turned white. A double-beam ultraviolet spectrophotometer (Specord 200 plus, Analytik Jena, Jena, Germany) was used to measure absorbance at 470, 645, and 663 nm. CHL and CAR were calculated using the following equations [28]:

$$\mathrm{C_a}\left(\mathrm{mg\ L^{-1}}\right) = 12.7\mathrm{A_{663}} - 2.69\mathrm{A_{645}} \tag{3}$$

$$\mathrm{C_b}\left(\mathrm{mg\ L^{-1}}\right) = 22.9\mathrm{A_{645}} - 4.68\mathrm{A_{663}} \tag{4}$$

$$\mathrm{C_{car}}\left(\mathrm{mg\ L^{-1}}\right) = \frac{1000\mathrm{A_{470}} - 3.27\mathrm{C_a} + 104\mathrm{C_b}}{229} \tag{5}$$

$$\mathrm{CHL}\left(\mathrm{g\ m^{-2}}\right) = \frac{(\mathrm{C_a} + \mathrm{C_b}) \times \mathrm{V}}{\mathrm{W} \times 1000} \times \mathrm{LFW} \tag{6}$$

$$\mathrm{CAR}\left(\mathrm{g\ m^{-2}}\right) = \frac{\mathrm{C_{car}} \times \mathrm{V}}{\mathrm{W} \times 1000} \times \mathrm{LFW} \tag{7}$$

where $\mathrm{C_a}$, $\mathrm{C_b}$, and $\mathrm{C_{car}}$ represent the chlorophyll a, chlorophyll b, and carotenoids, respectively. $\mathrm{A_W}$ is the absorbance in band w, V is the volume of the extraction solution (40 mL), W is the fresh weight of the filamentous leaves, and LFW (g m^{-2}) is the fresh weight of green leaves per unit ground area.

LNC and LCC: The dried leaves were ground and sieved through a 0.15 mm mesh. Powder samples of about 50 mg were wrapped using aluminum foil and placed in an elemental analyzer (Vario MACRO cube, Elementar, Germany) to determine the total nitrogen and carbon contents of rice leaves. LNC and LCC were calculated using the following equations:

$$\mathrm{LNC}\left(\mathrm{g\ m^{-2}}\right) = \mathrm{TN} \times \mathrm{LDW} \tag{8}$$

$$\mathrm{LCC}\left(\mathrm{g\ m^{-2}}\right) = \mathrm{TC} \times \mathrm{LDW} \tag{9}$$

where TN and TC are the total nitrogen content and total carbon content, respectively. LDW (g m^{-2}) is the dry weight of leaves per unit ground area.

3) Determination of Grain Yield: At maturity, rice grains within each subplot were manually harvested for grain yield determination. Rice grains were sun-dried, impurities removed, and weighed using an electronic balance. The grain yield was calculated at a standard water content of 13.5% and converted to tons per hectare (t $\cdot$ hm^{-2}).

C. Hyperspectral Image Preprocessing

The preprocessing workflow for hyperspectral images is shown in Fig. 2.

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TABLE II
SPECIFIC SPECIFICATIONS OF THE AIRCRAFT AND HYPERSPECTRAL CAMERA USED IN THIS STUDY

Aircraft		Hyperspectral camera	
Type	DJI Matrice 600 Pro	Type	Resonon Pika L
Aircraft weight	10 kg	Spectral range	400–1000 nm
Maximum take-off weight	15.5 kg	Spectral resolution	3.3 nm
Maximum flying height	2500 m	Spectral bandwidth	2.1 nm
Maximum flying speed	18 m s^{-1}	Spectral channels	281
Size (folded)	166.8 × 151.8 × 72.7 cm	Spatial pixels per line	900
		Frame rate	87 fps
		Focal length	17 mm
		FOV/IFOV	16.7°/0.71 mrad
		Data format	*.bil
		Weight	0.7 kg
		Size	12.5 × 10 × 5.3 cm

Note: FOV and IFOV are the field of view and the instantaneous field of view, respectively.

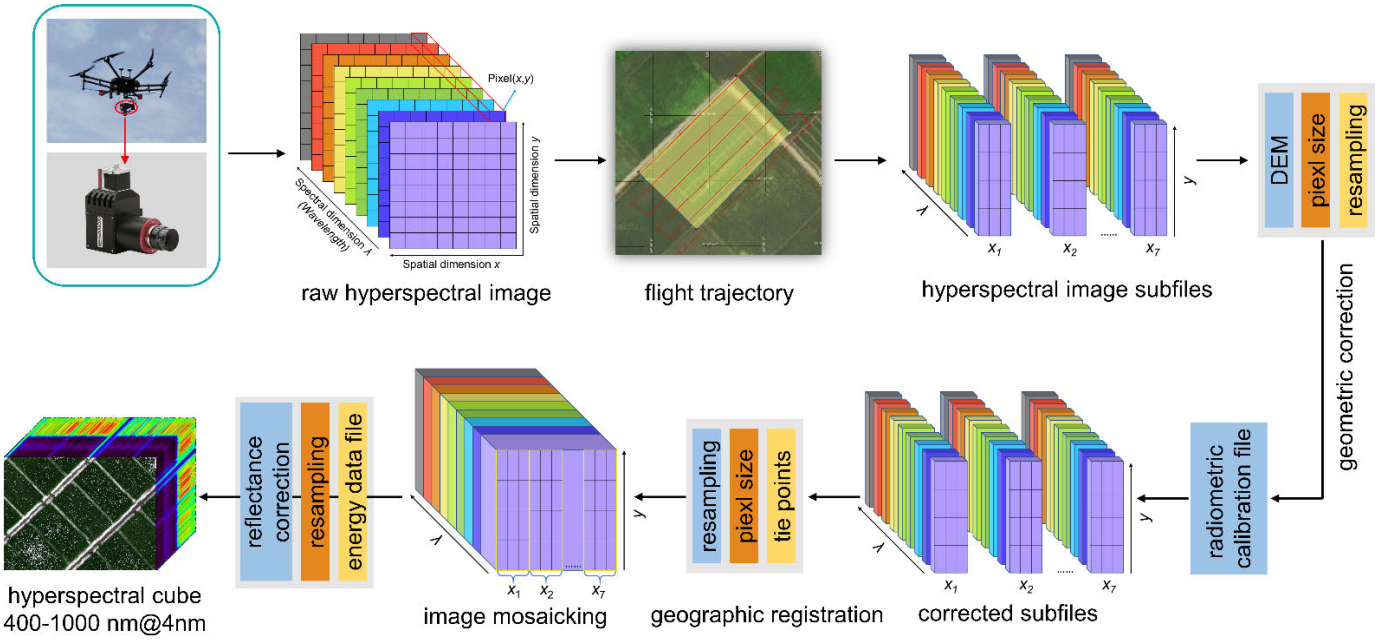


Fig. 2. Flowchart of hyperspectral image preprocessing.

1) *Data Segmentation:* Due to the large storage of raw files (~20 GB), flight trajectory information derived from POS data was used to spatially segment the raw image to improve processing efficiency.
2) *Geometric and Radiometric Correction:* The hyperspectral image subfiles were geometrically and radiometrically corrected by inputting a radiometric calibration file in MegaCube software (v2.7.0, IRIS, Inc., Beijing, China).
3) *Geographic Registration:* The hyperspectral image subfiles were imported into ArcMap software (v10.7, Esri Inc., CA, USA), and the "georeferencing" tool was used to perform geographic registration by matching tie points between adjacent subfiles.
4) *Image Mosaicking:* All hyperspectral image subfiles were mosaicked into a complete hyperspectral image in ENVI software (v5.3.1, Exelis Visual Information Solutions, Boulder, CO, USA).
5) *Hyperspectral Cube Generation:* The hyperspectral images obtained in step 4) were loaded into the MegaCube software and input into the energy data file to generate a hyperspectral cube with a spatial resolution of 4 cm and a spectral range of 400–1000 nm.
6) *Reflectance Correction:* The hyperspectral cube was resampled to obtain hyperspectral images with a spectral band interval of 4 nm in ENVI software. The linear empirical method was used to perform reflectance correction to convert the image energy data into spectral reflectance data.

D. Vegetation Indices

Two categories of VIs were adopted to estimate the rice CGMI: VIs with randomly combined bands and VIs with specific bands from previous studies. These VIs have been proven to be closely related to dry matter, water, photosynthetic

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TABLE III
DEFINITION AND EQUATIONS OF VIS USED IN THIS STUDY

Category	VIs	Equation	Reference
VIs with randomly combined bands	DVI	$R_{nir} - R_{red}$	[29]
	RVI	R_{nir}/R_{red}	
	NDVI	$(R_{nir} - R_{red})/(R_{nir} + R_{red})$	[30]
	GNDVI	$(R_{nir} - R_{green})/(R_{nir} + R_{green})$	[31]
	EVI	$2.5 \times [(R_{nir} - R_{red})/(1 + R_{nir} + 6 \times R_{red} - 7.5 \times R_{blue})]$	[32]
VIs with specific bands	$CI_{rededge}$	$(R_{770}/R_{710}) - 1$	[33]
	$CRI_{rededge}$	$[(R_{510})^{-1} - (R_{730})^{-1})] \times R_{770}$	[34]
	WI	R_{900}/R_{970}	[35]
	DCNI	$(R_{720} - R_{700})/(R_{700} - R_{670})/(R_{720} - R_{670} + 0.03)$	[36]
	PRI	$(R_{531} - R_{570})/(R_{531} + R_{570})$	[37]

Note: R, nir, red, green, and blue represent spectral reflectance, near-infrared band, red and red-edge band, green band, and blue band, respectively.

pigments, and nitrogen. Therefore, they were selected to comprehensively characterize the growth parameters associated with rice growth status. VIs with randomly combined bands included difference VI (DVI), ratio VI (RVI), normalized DVI (NDVI), green NDVI (GNDVI), and enhanced VI (EVI). DVI, RVI, and NDVI were calculated from all possible two-by-two combinations of red/red-edge bands (630–780 nm) and near-infrared (800–1000 nm) bands. GNDVI was calculated using all possible 2×2 combinations of green bands (518–572 nm) and near-infrared bands. EVI was calculated using all possible 3×3 combinations of the blue bands (400–498 nm), red/red-edge bands, and near-infrared bands. Five VIs with specific bands were calculated to develop the estimation models of rice CGMI, namely, red-edge chlorophyll index ($CI_{rededge}$), red-edge carotenoid index ($CRI_{rededge}$), water index (WI), double-peak canopy nitrogen index (DCNI), and photochemical reflectance index (PRI). Definitions and equations of VIs are listed in Table III.

E. Inversion Framework of CGMI

The general procedures of CGMI inversion and comprehensive growth assessment are as follows.

1) *Data Processing:* The Savitzky–Golay convolutional smoothing algorithm was adopted to weaken the high-frequency noise of spectral reflectance using a window size of 15 and a second-degree polynomial in Python 3.11. Two parameter settings can provide a suitable balance between noise reduction and preservation of spectral features [4]. The seven growth parameters were normalized to scale the data using the min–max normalization method.
2) *CGMIs Calculation:* $CGMI_{EWM}$, $CGMI_{CV}$, and $CGMI_{IFS}$ were calculated using the EWM, CV, and IFS methods, respectively.
3) *CGMIs Comparison:* The differences between the three CGMIs were compared with respect to seven growth parameters.
4) *CGMIs Estimation:* The feature selection algorithm was used to select the more important VIs, and the estimation models of CGMIs were developed using machine learning and deep learning algorithms.
5) *CGMIs Inversion:* The model performance of $CGMI_{EWM}$, $CGMI_{CV}$, and $CGMI_{IFS}$ was evaluated and compared to determine the optimal model to realize the comprehensive rice growth assessment. In this study, quadratic relationships of CGMIs with grain yield were quantified to determine the CGMI level classifications using a yield-decline-ratio-based thresholding strategy.

F. EWM Method

The EWM is an objective weighting technique based on the concept of entropy from information theory and has been widely applied in crop growth evaluations [22], [38], [39]. In information theory, entropy is used to quantify the uncertainty of a system; the greater the dispersion of an indicator, the higher its entropy value and the higher the amount of information it contains [40]. In crop growth evaluations, discrepancies in dimensionality, scales, and variability often exist among indicators, and direct weighting may lead to bias. EWM quantifies the entropy of each indicator to derive its redundancy (i.e., effective information), enabling a more objective and data-driven allocation of weights. The calculation procedure of EWM is as follows.

1) Let $X = \{x_{ij}\} \in R^{n \times m}$ be the original data matrix with n indicators and m samples. In this study, seven growth parameters positively correlated with grain yield were treated as positive indicators. To eliminate the effects of dimensionality, the min–max normalization method was applied to convert raw values x_{ij} into dimensionless values z_{ij}

$$z_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)}, \quad i = 1, 2, 3, \ldots, n; \quad j = 1, 2, 3, \ldots, m \tag{10}$$

where z_{ij} is the normalized value of sample x_{ij}. x_{ij} is the value of the sample i of the indicator j. $\max(x_j)$ and $\min(x_j)$ are the maximum and minimum values of indicator j, respectively. n and m denote the number of samples and indicators, respectively.
2) The p_{ij} is the proportion of the normalized value of sample i for indicator j. This proportion matrix was

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used to calculate entropy and reflects the distribution characteristics of each indicator

$$p_{ij} = \frac{z_{ij}}{\sum_{i=1}^{n} z_{ij}} \tag{11}$$

where n denotes the number of samples.

3) The entropy value e_j of each indicator was calculated to assess the distribution uniformity of data

$$e_j = -\frac{1}{\ln(n)} \sum_{i=1}^{n} p_{ij} \ln\left(p_{ij}\right) \tag{12}$$

where $(1/\ln(n))$ denotes the normalization coefficient, ensuring that the entropy value is constrained within the range of 0–1, facilitating comparison among indicators. $\ln(p_{ij})$ denotes the uncertainty associated with the distribution of indicator values.

4) The information redundancy d_j quantifies the degree of variation for indicator j across samples. Greater variability indicates stronger discriminative ability and justifies a higher objective weight

$$d_j = 1 - e_j. \tag{13}$$

5) The objective weight of each indicator was obtained by normalizing the redundancy values

$$w_j = \frac{d_j}{\sum_{j=1}^{m} d_j} \tag{14}$$

where w_j is the objective weight of indicator j, m denotes the number of indicators.

6) The $\mathrm{CGMI_{EWM}}$ values were calculated by adding the product of the normalized values of each indicator and the objective weights

$$\mathrm{CGMI_{EWM}} = \sum_{j=1}^{m} z_{ij} \times w_j. \tag{15}$$

G. CV Method

The CV method is an objective weight technique based on the degree of data dispersion, which is quantified by the ratio of standard deviation to mean. The core principle of the CV method is that the greater the variability of an indicator across samples, the stronger its discriminative ability, and the higher the weight it should receive [41]. In crop growth evaluations, the CV method effectively emphasizes the importance of highly variable indicators and serves as a commonly used data-driven weight method. The calculation procedure of the CV method is as follows.

1) As in Step 1 of the EWM method, the min–max normalization method was used to ensure all indicators were dimensionless and positively oriented using (10).
2) The cv_j of each indicator was calculated using the ratio of standard deviation to mean

$$cv_j = \frac{s_j}{\bar{x}_j}, \quad j = 1, 2, 3, \ldots, m \tag{16}$$

where cv_j denotes the CV of indicator j. s_j is the standard deviation of indicator j, and $\bar{x}_j$ is the mean of indicator j. m is the number of indicators.

3) The CV values of all indicators were normalized to derive the objective weight w_j for indicator j

$$w_j = \frac{cv_j}{\sum_{j=1}^{m} cv_j} \tag{17}$$

where w_j represents the weight of the indicator j.

4) The $\mathrm{CGMI_{CV}}$ values were calculated by adding the product of the normalized values of each indicator and the objective weights

$$\mathrm{CGMI_{CV}} = \sum_{j=1}^{m} z_{ij} \times w_j \tag{18}$$

where z_{ij} is the normalized value of the sample i for the indicator j.

H. IFS Method

Different growth parameters exhibit varying capabilities to characterize crop growth status, and such capabilities may fluctuate due to noise, data distribution, and data attributes. The traditional EWM and CV methods determine objective weights based on a single attribute, either entropy or variability, without considering discriminative power across categories and stability within groups. These methods also lack an adaptive mechanism for weight adjustment. Therefore, this study proposed an IFS method in which pseudo-labels are created using unsupervised clustering. The method fused the discriminative ability of each indicator, measured by information gain ratio (IGR), with its stability, measured by the ratio of within-cluster variance (WCV). A tunable parameter α was introduced to adaptively balance these two components, and the optimal α^* was determined using a cross-validation method. This framework enabled the construction of a more stable and reliable rice CGMI. The workflow of the IFS method is shown in Fig. 3. The calculation procedure of the IFS method is as follows.

1) All indicators were positively normalized using (10).
2) The K-means clustering algorithm was employed to divide all samples into K growth categories, serving as pseudo-labels for subsequent analysis of indicator stability and discriminative ability. In this study, the number of clusters was preset to K = 3 based on agronomic considerations rather than statistical optimization. Specifically, rice growth status was classified into three levels, namely, weak growth, suitable growth, and excessive growth, which are commonly used to characterize crop growth status in practical field management

$$\begin{cases} \min \sum_{k=1}^{K} \sum_{z_i \in \mathrm{C}_k}^{\|z_i - \mu_k\|^2} \\ \mathrm{C}_i \in \{1, 2, 3, \ldots, K\}, \quad i = 1, 2, 3, \ldots, n \end{cases} \tag{19}$$

where C_k denotes the kth cluster, z_i is the normalized feature vector of sample i, μ_k represents the centroid of cluster k, and K is the number of clusters. C_i denotes the cluster label assigned by the K-means clustering algorithm, and n denotes the number of samples.

3) The WCV and total variance of indicator j were calculated, and the stability index (Stab) was defined to

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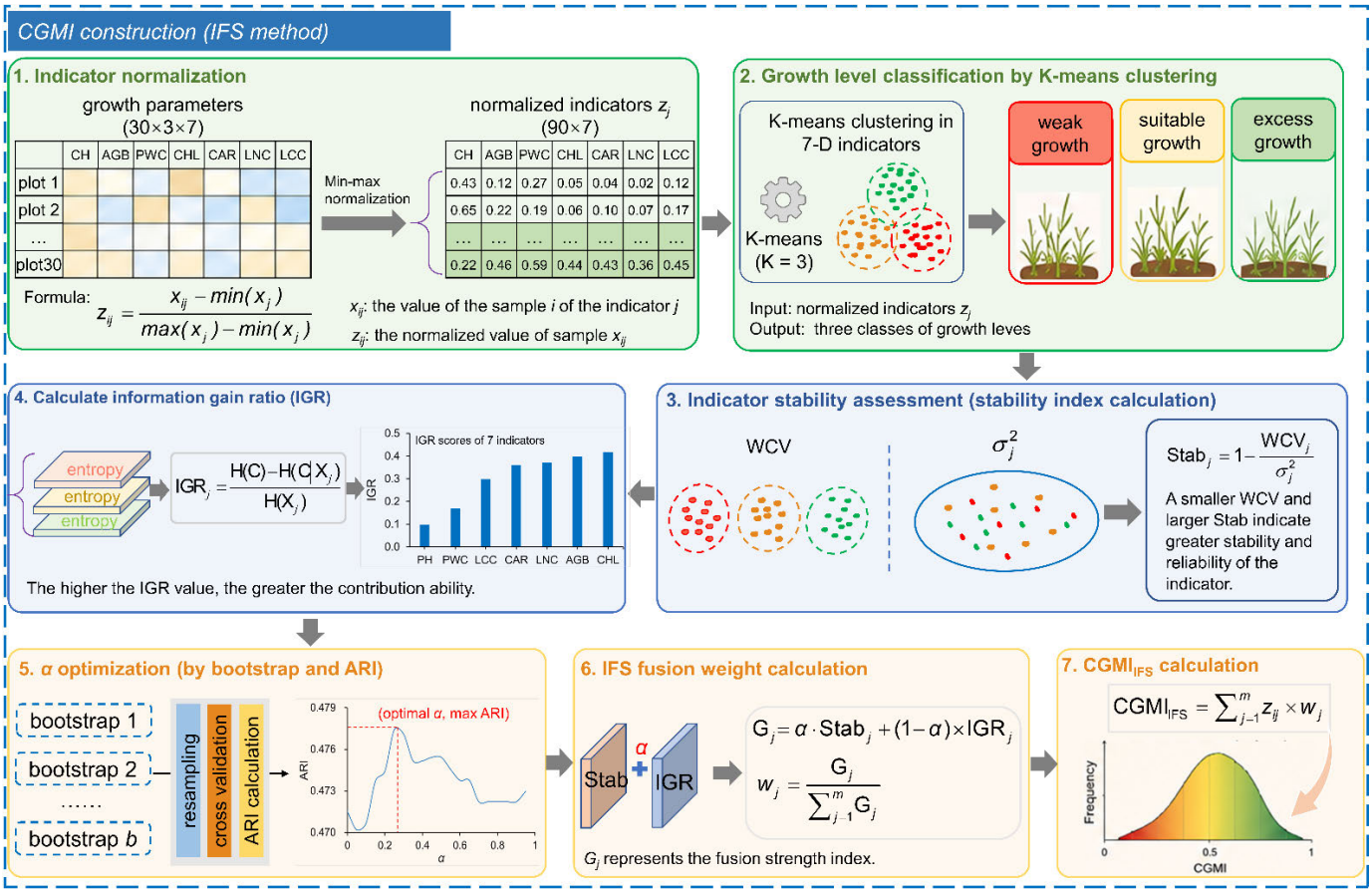


Fig. 3. Workflow of the IFS method.

characterize the stability and reliability of the indicator across different cluster categories

$$\mathrm{WCV}_j = \sum_{k=1}^{K} \frac{nk}{n} \times \sigma_{jk}^2 \tag{20}$$

$$\mathrm{Stab}_j = 1 - \frac{\mathrm{WCV}_j}{\sigma_j^2} \tag{21}$$

where nk is the number of samples in cluster k, σ_{jk}^2 is the variance of indicator j in cluster k, and σ_j^2 is the total variance of indicator j. A smaller WCV and larger Stab indicate greater stability and reliability of the indicator.

4) Each indicator was discretized using quantile bins, and the IGR of the clustering labels was calculated as follows.

a) Informational entropy after discretization

$$\mathrm{H}(\mathrm{X}_j) = -\sum_{r=1}^{R} \mathrm{P}(\mathrm{D}_{jr}) \log_2 (\mathrm{D}_{jr}) \tag{22}$$

where $\mathrm{H}(\mathrm{X}_j)$ is the entropy of indicator j, $\mathrm{P}(\mathrm{D}_{jr})$ is the probability of samples falling into interval r, D_{jr} denotes the rth discretized interval of indicator X_j, and R is the number of intervals after discretization.

b) Informational entropy of cluster labels

$$\mathrm{H}(\mathrm{C}) = -\sum_{k=1}^{K} \mathrm{P}(\mathrm{C}_k) \log_2 \mathrm{P}(\mathrm{C}_k) \tag{23}$$

where $\mathrm{H}(\mathrm{C})$ is the entropy of cluster labels, $\mathrm{P}(\mathrm{C}_k)$ is the probability that a sample belongs to cluster k, C_k denotes the kth cluster, and K is the number of clusters.

c) Conditional entropy of discretized indicator X_j

$$\begin{cases} \mathrm{H}(\mathrm{C}|\mathrm{X}_j) = \sum_{r=1}^{R} \mathrm{P}(\mathrm{D}_{jr}) \times \mathrm{H}(\mathrm{C}|\mathrm{X}_j \in \mathrm{D}_{jr}) \\ \mathrm{H}(\mathrm{C}|\mathrm{X}_j \in \mathrm{D}_{jr}) = -\sum_{k=1}^{K} \mathrm{P}(\mathrm{C}_k|\mathrm{D}_{jr}) \\ \quad \times \log_2 \mathrm{P}(\mathrm{C}_k|\mathrm{D}_{jr}) \end{cases} \tag{24}$$

where $\mathrm{H}(\mathrm{C}|\mathrm{X}_j)$ is the conditional entropy of cluster labels given indicator X_j, and $\mathrm{H}(\mathrm{C}|\mathrm{X}_j \in \mathrm{D}_{jr})$ is the entropy of cluster labels within interval D_{jr}. $\mathrm{P}(\mathrm{C}_k|\mathrm{D}_{jr})$ is the conditional probability that a sample belongs to cluster k within interval D_{jr}.

d) IGR

$$\mathrm{IGR}_j = \frac{\mathrm{H}(\mathrm{C}) - \mathrm{H}(\mathrm{C}|\mathrm{X}_j)}{\mathrm{H}(\mathrm{X}_j)} \tag{25}$$

where a higher IGR indicates that indicator j contributes more to reducing clustering uncertainty.

5) A fusion coefficient $\alpha \in [0,1]$ was introduced to combine indicator stability and discriminability

$$\mathrm{G}_j = \alpha \cdot \mathrm{Stab}_j + (1 - \alpha) \times \mathrm{IGR}_j \tag{26}$$

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TABLE IV
STATISTICAL RESULTS OF THE CGMIS AND GRAIN YIELD FOR THE TRAINING SET AND TEST SET

Growth stages	Parameters	Training set				Test set			
		Min	Max	Mean	S.td	Min	Max	Mean	S.td
Tillering stage	$CGMI_{EWM}$	0.056	0.928	0.404	0.199	0.081	0.892	0.407	0.180
$n = 90$	$CGMI_{CV}$	0.268	0.970	0.538	0.160	0.311	0.952	0.555	0.138
	$CGMI_{IFS}$	0.040	0.956	0.393	0.207	0.082	0.908	0.413	0.182
Jointing stage	$CGMI_{EWM}$	0.044	0.931	0.390	0.209	0.056	0.867	0.370	0.173
$n = 90$	$CGMI_{CV}$	0.218	0.966	0.507	0.178	0.275	0.991	0.536	0.151
	$CGMI_{IFS}$	0.029	0.946	0.378	0.221	0.067	0.952	0.395	0.189
Full-heading stage	$CGMI_{EWM}$	0.074	0.810	0.408	0.176	0.118	0.856	0.473	0.140
$n = 90$	$CGMI_{CV}$	0.254	0.907	0.554	0.157	0.278	0.943	0.594	0.118
	$CGMI_{IFS}$	0.069	0.861	0.433	0.187	0.122	0.836	0.478	0.141
Maturity stage	Grain yield	6.282	10.068	8.452	0.965	8.272	10.553	9.316	0.651
$n = 30$	($t\ hm^{-2}$)								

where G_j represents the fusion strength index, and α is the fusion coefficient, which was searched from 0 to 1 with an interval of 0.05.

Then, the fusion weight was obtained by normalizing G_j

$$w_j = \frac{G_j}{\sum_{j=1}^{m} G_j} \tag{27}$$

where w_j represents the fusion weight of indicator j.

6) The $CGMI_{IFS}$ was calculated as the weighted sum of all normalized indicators

$$CGMI_{IFS} = \sum_{j=1}^{m} z_{ij} \times w_j \tag{28}$$

where z_{ij} is the normalized value of sample i for indicator j.

7) In this study, a pseudo-supervised cross-validation framework was designed to determine the optimal fusion coefficient α^*.
   a) Bootstrap resampling with cross validation was conducted with 30 iterations.
   b) For each resampled subset, Steps (2)–(6) were repeated to calculate $CGMI_{IFS}$ using the candidate α.
   c) K-means was applied to predict the CGMI-based category.
   d) The adjusted rand index (ARI) was calculated using predicted and original clustering labels

$$ARI^{(b)}(\alpha) = ARI\left(KMeans\left(Z^{(b)}\right), Q\left(CGMI^{(b)}(\alpha), K\right)\right) \tag{29}$$

   where $ARI^{(b)}(\alpha)$ represents the clustering consistency score of α under the bth cross-validation iterations, $Z^{(b)}$ is the dataset obtained from the bth bootstrap resampling, Q represents the quantile grouping operation, $CGMI^{(b)}(\alpha)$ is the CGMI calculated using the fusion coefficient α, and K is the number of clusters.
   e) The average ARI over all bootstrap iterations was calculated

$$\overline{ARI}(\alpha) = \frac{1}{B}\sum_{b=1}^{B} ARI^{(b)}(\alpha),\ b = 1, 2, 3, \ldots, B \tag{30}$$

   where B represents the number of bootstrap resampling iterations, and $\overline{ARI}$ represents the average value of ARI.
   f) The fusion factor that corresponds to the maximum $\overline{ARI}$ was defined as the optimal fusion factor α^*

$$\alpha^* = \arg\max \overline{ARI}(\alpha). \tag{31}$$

I. Model Evaluation of CGMI

The workflow for developing the CGMIs estimation models is shown in Fig. 4. The dataset from 2022 was used as the training set, and the dataset from 2023 was used as the test set to assess model robustness and generalizability. Statistical results of the training and test sets are presented in Table IV. In this study, the seven growth parameters, grain yield, and hyperspectral data were cleaned to check for missing values and remove outliers. Three machine learning algorithms were used to develop CGMI estimation models, including partial least-squares regression (PLSR), random forest (RF) regression, and support vector regression (SVR). Considering that some VIs may contain overlapping spectral information, the permutation importance method was employed to identify the most informative variables during model development. This method evaluates the contribution of each feature by measuring the decline in model performance when the feature values are randomly permuted. A greater performance drop indicates a higher dependency of the model on that feature [42]. In addition, a multilayer perceptron (MLP) algorithm was introduced as a deep-learning model to further evaluate the applicability of the CGMIs. To ensure a fair comparison among different modeling approaches, the MLP model used the VIs subsets selected by the permutation importance method for the corresponding PLSR, RF, and SVR models. Moreover, all

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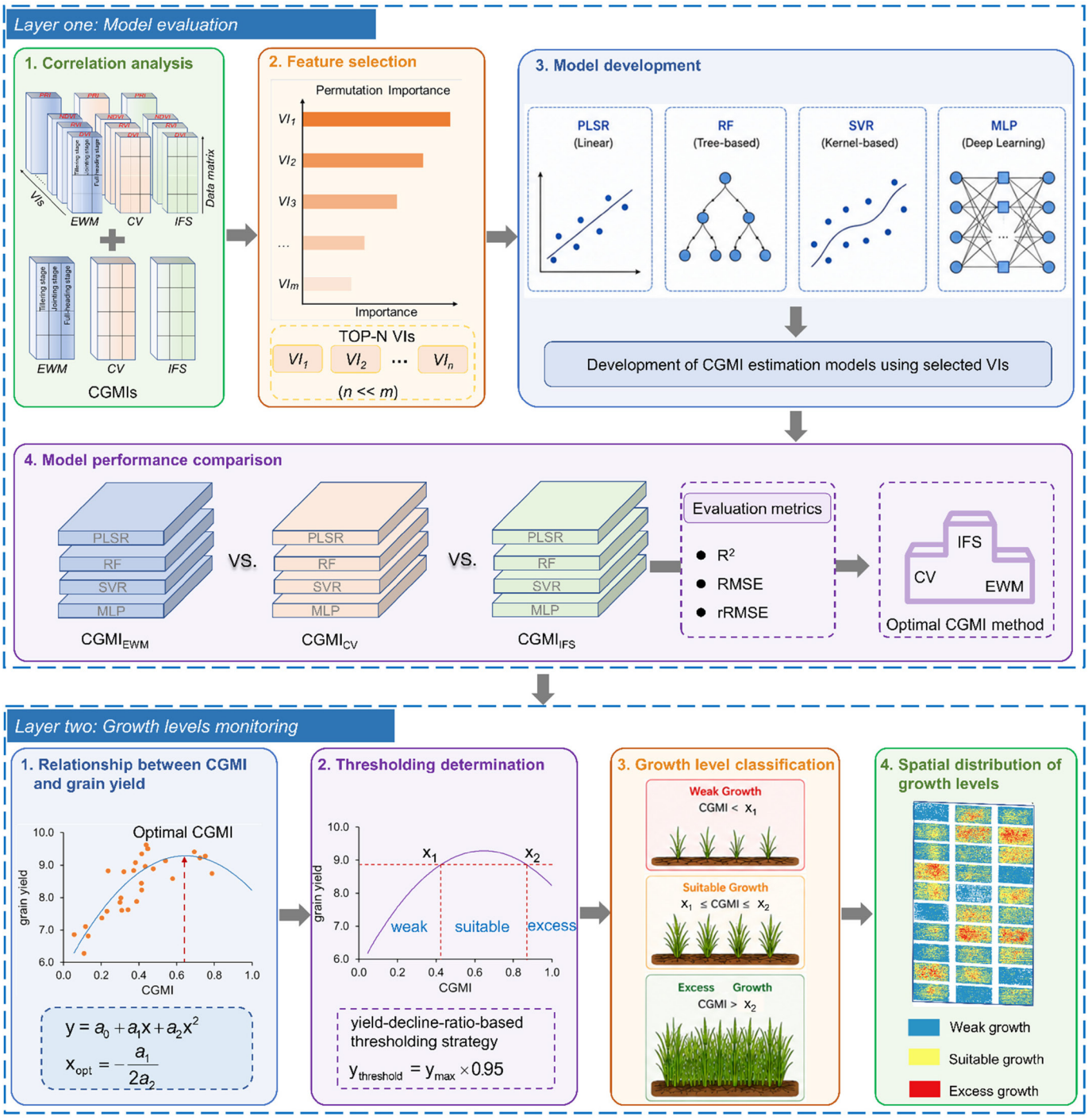


Fig. 4. Workflow for the development of CGMIs estimation models and growth levels assessment.

regression algorithms are generally robust to multicollinearity among predictor variables. All machine-learning models were implemented using the scikit-learn module in Python 3.11, whereas the MLP model was implemented using PyTorch. The grid search method was used to optimize the hyperparameters of the regression models. The optimal hyperparameter results and related information of PLS, RF, and SVR were presented in Tables A.I–A.III. The hyperparameters configured for the MLP in deep learning are shown in Table A.IV. In this study, all VIs were standardized, and the top seven most important features by permutation importance were selected to develop regression models. A limited number of highly informative features was retained to reduce information redundancy, alleviate potential multicollinearity among VIs, and avoid unnecessary model complexity while preserving the most relevant spectral information for CGMI estimation. Three metrics were adopted to evaluate model performance, namely, the coefficient of determination (R^2), root-mean-square error

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TABLE V
WEIGHT COEFFICIENT OF GROWTH PARAMETERS FOR DIFFERENT CGMIS CALCULATION METHODS

	Tillering stage			Jointing stage			Full-heading stage		
	$CGMI_{EWM}$	$CGMI_{CV}$	$CGMI_{IFS}$	$CGMI_{EWM}$	$CGMI_{CV}$	$CGMI_{IFS}$	$CGMI_{EWM}$	$CGMI_{CV}$	$CGMI_{IFS}$
CH	0.173	0.077	0.047	0.136	0.073	0.087	0.124	0.088	0.084
AGB	0.131	0.137	0.176	0.127	0.141	0.144	0.087	0.111	0.145
PWC	0.061	0.009	0.081	0.061	0.009	0.031	0.155	0.017	0.027
CHL	0.168	0.207	0.180	0.191	0.219	0.189	0.195	0.231	0.208
CAR	0.165	0.197	0.175	0.190	0.209	0.195	0.183	0.222	0.205
LNC	0.171	0.212	0.170	0.149	0.187	0.179	0.146	0.184	0.171
LCC	0.130	0.161	0.170	0.146	0.161	0.175	0.110	0.147	0.160

(RMSE), and relative RMSE (rRMSE), which were calculated using the following equations:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(y_i - y_i'\right)^2}{\sum_{i=1}^{n} \left(y_i - \bar{y}\right)^2} \tag{32}$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} \left(y_i - y_i'\right)^2}{n}} \tag{33}$$

$$\text{rRMSE} = \frac{\text{RMSE}}{\bar{y}} \times 100\% \tag{34}$$

where y_i and y_i' indicate the measured values and predicted values of sample i, respectively. $\bar{y}$ indicates the mean measured values, and n is the number of samples for the training or test set. The performance of rice CGMI estimation models developed by EWM, CV, and IFS methods was compared. The remotely sensed maps of growth levels were drawn to assess the growth status of rice. To further evaluate the robustness and transferability of the proposed IFS method, an independent dataset collected from a different experimental site under different management conditions was used for external validation. Please see the supplementary materials for experimental details.

J. Yield-Decline-Ratio-Based Thresholding Strategy

In this study, a growth level classification method was established using an agronomically driven yield-decline-ratio-based thresholding strategy (see Fig. 4). First, the quadratic relationship between CGMI and grain yield at the tillering, jointing, and full-heading stages was determined, which can be expressed as (35). Second, the coefficients (a_1 and a_2) of the quadratic equation were used to calculate the optimal CGMI (x_{opt}), that is, the CGMI corresponding to the maximum value of grain yield. The optimal CGMI was calculated using (36). Third, the maximum grain yield (y_{max}) was calculated by inputting $CGMI_{opt}$ into the quadratic equation. The threshold coefficient of grain yield was set to 0.95, which was calculated from the ratio of the local high-yield production standard (9 $t \cdot hm^{-2}$) to the maximum grain yield. The threshold coefficient was consistent with the commonly adopted criterion of approximately 95% of the maximum attainable yield for defining optimal crop production conditions [43]. The grain yield threshold ($y_{threshold}$) was calculated using (37). Finally, the rooting formula of the quadratic equation was used to calculate two solutions (x_1 and x_2) as the lower (minimum x_1) and upper (maximum x_2) limits for the growth level classification, as in (38)

$$y = a_0 + a_1 x + a_2 x^2 \tag{35}$$

$$x_{opt} = -\frac{a_1}{2a_2} \tag{36}$$

$$y_{threshold} = y_{max} \times 0.95 \tag{37}$$

$$x_{level} = \begin{cases} x < x_1, & \text{weak growth} \\ x_1 \le x \le x_2, & \text{suitable growth} \\ x > x_2, & \text{excessive growth} \end{cases} \tag{38}$$

where y represents the grain yield, and x represents the CGMI. a_0, a_1, and a_2 represent the intercept, linear term coefficient, and quadratic term coefficient, respectively. x_{level} represents the growth level classification, including weak growth, suitable growth, and excessive growth.

III. RESULTS

A. Relationships Between Growth Parameters and CGMIs

Entropy weights, coefficients of variation, and information fusion were used to determine the weight coefficients of the growth parameters in this study (see Table V). The changes and differences in the three CGMIs with elevated growth parameters were comparatively analyzed to test the responsiveness of the comprehensive growth indicators to the growth parameters, as shown in Fig. 5. The increasing trends of CH and PWC in rice were unrelated to the changes in CGMIs, and the three CGMIs changed without regularity at the tillering, jointing, and full-heading stages. The weight coefficients of CH and PWC for $CGMI_{CV}$ and $CGMI_{IFS}$ were examined to be lower than those of other growth parameters. The trends of the other growth parameters and the three CGMIs were generally consistent and showed positive correlations. However, the weight coefficients for $CGMI_{EWM}$, $CGMI_{CV}$, and $CGMI_{IFS}$ were lower on the AGB and LCC than on the CHL, CAR, and LNC, which may lead to greater fluctuation. The values of $CGMI_{CV}$ were higher than those of $CGMI_{EWM}$ and $CGMI_{IFS}$, but it is susceptible to suffering from extreme values with drastic changes in sample fluctuations. Although the trend of the $CGMI_{EWM}$ was stable, it was slightly lagging, which made it difficult to fully reflect the rapid growth trend of each growth parameter. The IFS method exhibited the least fluctuation and the most stable response trend across multiple growth parameters, and better follows the changes in growth parameters, suggesting a better fusion adaptation.

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TABLE VI
CORRELATION BETWEEN GROWTH PARAMETERS AND CGMIS

	Tillering stage			Jointing stage			Full-heading stage		
	$CGMI_{EWM}$	$CGMI_{CV}$	$CGMI_{IFS}$	$CGMI_{EWM}$	$CGMI_{CV}$	$CGMI_{IFS}$	$CGMI_{EWM}$	$CGMI_{CV}$	$CGMI_{IFS}$
CH	0.561^{**}	0.399^{**}	0.454^{**}	0.711^{**}	0.640^{**}	0.703^{**}	0.630^{**}	0.567^{**}	0.572^{**}
AGB	0.945^{**}	0.961^{**}	0.912^{**}	0.919^{**}	0.934^{**}	0.891^{**}	0.779^{**}	0.826^{**}	0.790^{**}
PWC	0.713^{**}	0.679^{**}	0.729^{**}	0.396^{**}	0.367^{**}	0.390^{**}	0.398^{**}	0.305^{**}	0.332^{**}
CHL	0.961^{**}	0.990^{**}	0.959^{**}	0.977^{**}	0.983^{**}	0.966^{**}	0.979^{**}	0.970^{**}	0.949^{**}
CAR	0.960^{**}	0.989^{**}	0.956^{**}	0.976^{**}	0.981^{**}	0.962^{**}	0.978^{**}	0.970^{**}	0.948^{**}
LNC	0.948^{**}	0.979^{**}	0.954^{**}	0.971^{**}	0.980^{**}	0.959^{**}	0.880^{**}	0.914^{**}	0.881^{**}
LCC	0.964^{**}	0.987^{**}	0.947^{**}	0.965^{**}	0.979^{**}	0.944^{**}	0.838^{**}	0.890^{**}	0.856^{**}

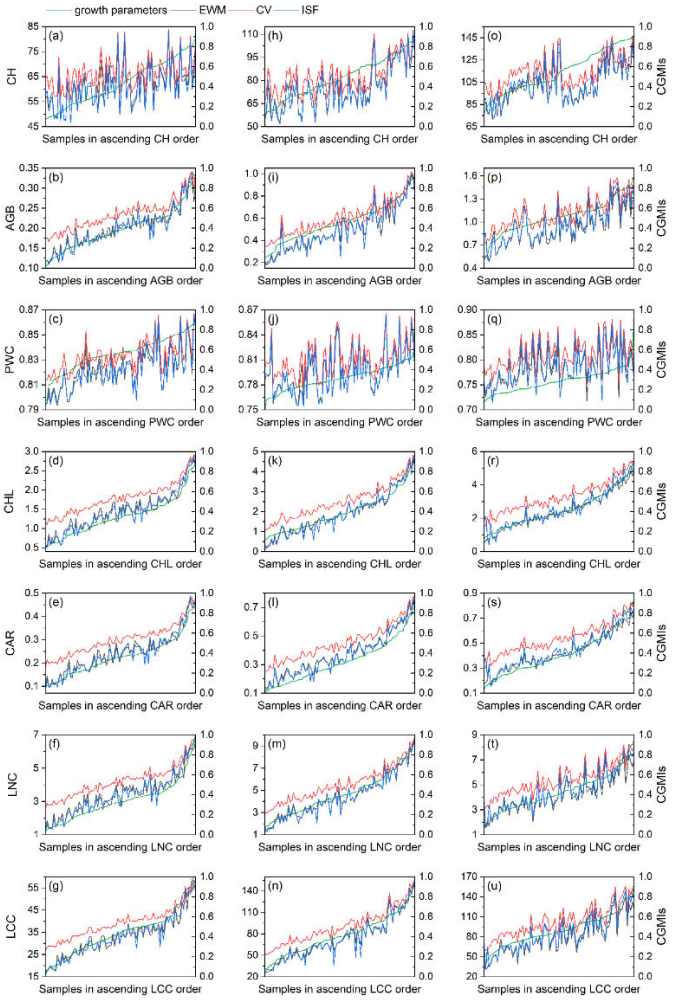


Fig. 5. Relationship between seven growth parameters and three CGMIs. (a)–(g), (h)–(n), and (o)–(u) Tillering, jointing, and full-heading stages, respectively.

The correlation coefficients between growth parameters and CGMIs are shown in Table VI. All growth parameters achieved the significance level ($p < 0.01$) with CGMIs. CH and PWC exhibited weaker sensitivity than the other five traits, which was consistent with the results in Fig. 5. CHL demonstrated the most robust sensitivity (r = 0.949–0.990) for each growth stage, which indicated that canopy chlorophyll contributed more to CGMIs.

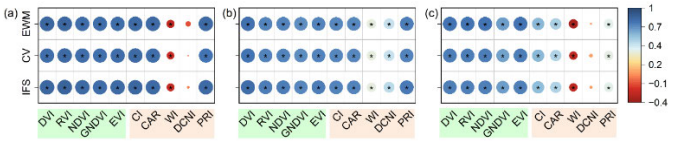


Fig. 6. Relationship between VIs and three CGMIs. (a)–(c) Tillering, jointing, and full-heading stages, respectively. The green and red areas are VIs with randomly combined bands and VIs with specific bands, respectively.

B. Correlation Analysis Between VIs and CGMIs

The Pearson correlations between the VIs and the three CGMIs were explored, and the results are displayed in Fig. 6. At the tillering stage, the RVI of the randomly combined bands had the strongest and significant correlation with $CGMI_{EWM}$, $CGMI_{CV}$, and $CGMI_{IFS}$, with correlation coefficients of 0.853, 0.842, and 0.863, respectively. At the jointing and full-heading stages, the DVI showed the strongest correlation with $CGMI_{EWM}$, $CGMI_{CV}$, and $CGMI_{IFS}$, with correlation coefficients of 0.811, 0.803, and 0.818 (jointing stage) and 0.803, 0.775, and 0.767 (full-heading stage), respectively. For VIs with specific bands, WI and DCNI showed weaker correlation with the three CGMIs than the other VIs. At the tillering and jointing stages, $CI_{rededge}$ and PRI had the strongest correlation with $CGMI_{IFS}$, with correlation coefficients of 0.857 and 0.778, respectively. The best correlation of $CGMI_{EWM}$ was found to be at $CI_{rededge}$ for the full-heading stage with a correlation coefficient of 0.617. The VIs of the randomly combined bands were more sensitive than the VIs with specific bands for monitoring the comprehensive growth status. The correlation coefficients between CGMIs and the VIs showed a decreasing trend as the growth stage progressed, with $CGMI_{IFS}$ correlating better than $CGMI_{EWM}$ and $CGMI_{CV}$ at the tillering and jointing stages, but $CGMI_{EWM}$ correlated best with the VIs at the full-heading stage.

C. Estimation Models of CGMIs Using Hyperspectral VIs

The permutation importance method was used to select the seven important features for CGMIs estimation models. The feature ranking and selected VIs are shown in Fig. 7. PLSR, RF, SVR, and MLP regression algorithms were used to develop the rice growth monitoring models and to compare the training accuracies of the $CGMI_{EWM}$, $CGMI_{CV}$, and $CGMI_{IFS}$ estimation models, as shown in Table VII. The best training results of MLP were obtained using the VIs subset selected

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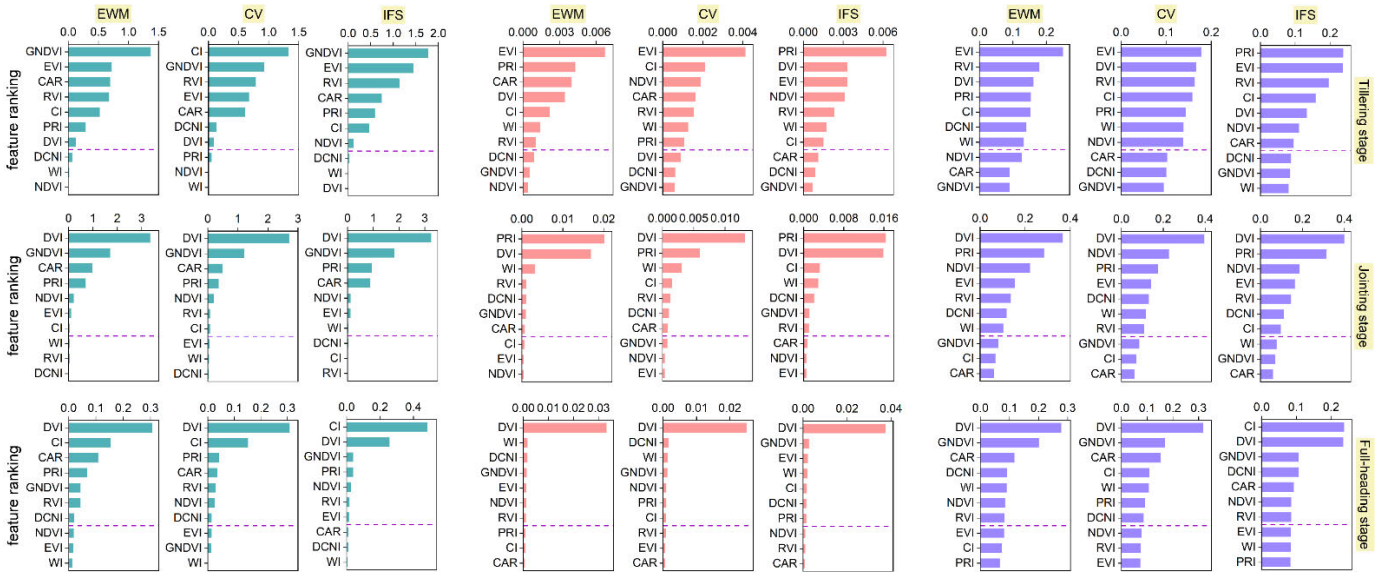


Fig. 7. Feature ranking and VIs selection using the permutation importance method. The blue, red, and purple represent PLSR, RF, and SVR, respectively. The top seven VIs above the dotted line were selected to establish the CGMIs estimation model.

TABLE VII
TRAINING RESULTS OF CGMIS ESTIMATION MODELS BASED ON THE SELECTED VIS

		PLSR			RF			SVR			PLSR-MLP		
		R^2	RMSE	rRMSE	R^2	RMSE	rRMSE	R^2	RMSE	rRMSE	R^2	RMSE	rRMSE
Tillering stage	$CGMI_{EWM}$	0.743	0.100	24.759	0.910	0.060	14.856	0.744	0.100	24.759	0.834	0.081	19.976
	$CGMI_{CV}$	0.718	0.084	15.605	0.904	0.049	9.103	0.710	0.085	15.791	0.717	0.084	15.691
	$CGMI_{IFS}$	0.786	0.055	13.990	0.941	0.032	8.140	0.851	0.059	15.007	0.940	0.050	12.799
Jointing stage	$CGMI_{EWM}$	0.699	0.114	29.207	0.927	0.056	14.347	0.693	0.115	29.463	0.807	0.092	23.448
	$CGMI_{CV}$	0.671	0.102	20.133	0.814	0.076	15.001	0.671	0.102	20.133	0.688	0.099	19.519
	$CGMI_{IFS}$	0.730	0.074	19.584	0.944	0.052	13.761	0.724	0.076	20.113	0.894	0.071	18.895
Full-heading stage	$CGMI_{EWM}$	0.654	0.103	25.233	0.790	0.080	19.599	0.685	0.098	24.008	0.691	0.097	23.861
	$CGMI_{CV}$	0.605	0.098	17.698	0.761	0.077	13.906	0.617	0.097	17.517	0.638	0.094	17.002
	$CGMI_{IFS}$	0.716	0.070	16.156	0.817	0.051	11.771	0.751	0.073	16.848	0.857	0.070	15.814

by PLSR (PLSR-MLP), while the training results of MLP obtained using the feature subsets selected by RF and SVR are shown in Table A.V. The values of R^2 and RMSE were often confounded by the magnitude and the attributes of the raw data, whereas rRMSE was able to standardize the error of models and facilitated the comparison of the model errors of the comprehensive growth indicators calculated by the different methods. For example, all estimation models of $CGMI_{EWM}$ had higher R^2 than those of $CGMI_{CV}$ for three growth stages, but their RMSE and rRMSE values were always larger than those of $CGMI_{CV}$. Even though the statistical results of $CGMI_{CV}$ were higher than those of $CGMI_{EWM}$ for three growth stages (see Table IV). The training accuracy of CGMI estimation models based on the IFS method was always better than that of the EWM and CV methods. Particularly, the RF regression algorithm exhibited the best training accuracy of $CGMI_{IFS}$ at the tillering, jointing, and full-heading stages with rRMSE values of 8.140%, 13.761%, and 11.771%, respectively.

D. Test of Estimation Models

The test set was used to validate the generalizability of the $CGMI_{EWM}$, $CGMI_{CV}$, and $CGMI_{IFS}$ estimation models, as shown in Fig. 8. The test results of MLP obtained using the VIs subsets selected by PLSR, RF, and SVR are shown in Table A.VI. Similar to the training accuracy, although the R^2 of the $CGMI_{EWM}$ estimation models was higher than that of $CGMI_{CV}$, the RMSE and rRMSE values of $CGMI_{CV}$ were lower than those of $CGMI_{EWM}$. The EWM method was less suitable for monitoring the comprehensive growth status compared with the CV and IFS methods. The $CGMI_{IFS}$ estimation models had the strongest test accuracy for the four regression algorithms over the three growth stages. Among them, the RF regression algorithm of $CGMI_{IFS}$ at the tillering, jointing, and full-heading stages showed the lowest test error, with rRMSE values of 7.275%, 11.364%, and 9.318%, respectively. Therefore, the RF estimation model of $CGMI_{IFS}$ demonstrated better robustness and adaptability, and IFS can be regarded as a preferred method for monitoring the comprehensive growth status of rice.

E. Relationships Between Grain Yield and CGMIs

The relationships between CGMIs and grain yield were analyzed, as shown in Fig. 9(a)–(c). The results showed that there was a quadratic relationship between CGMIs and grain

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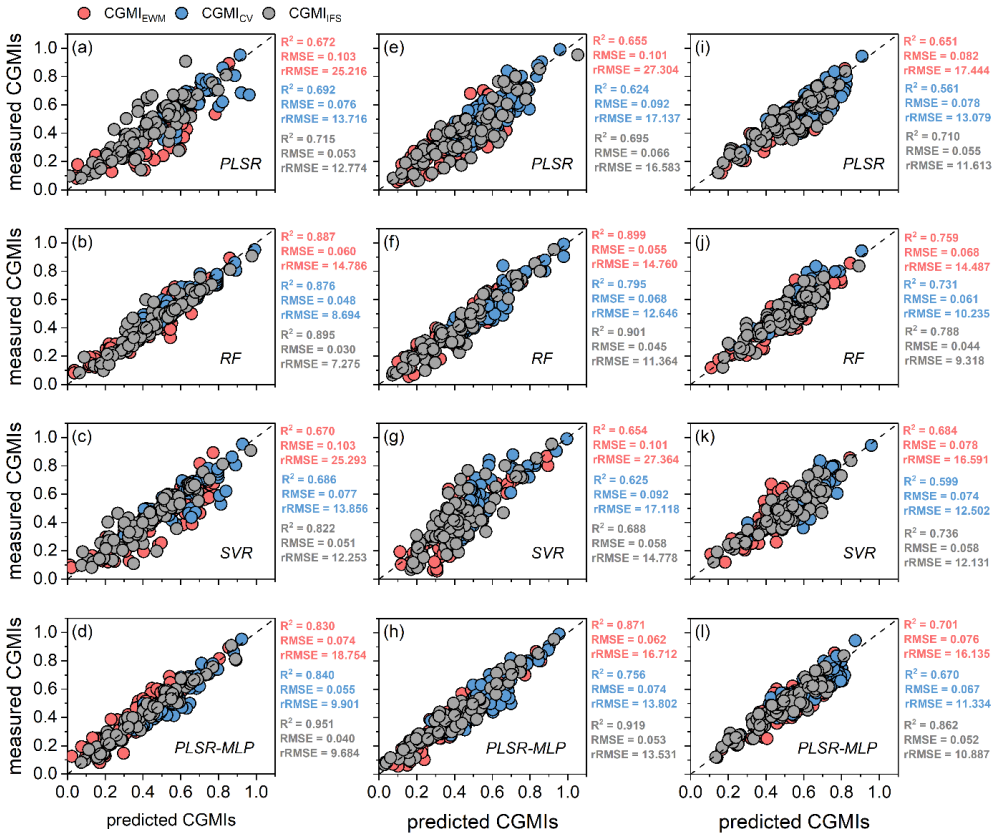


Fig. 8. Test accuracy of rice CGMIs estimation models based on EWM, CV, and IFS methods. (a)–(d), (e)–(h), and (i)–(l) Tillering, jointing, and full-heading stages, respectively.

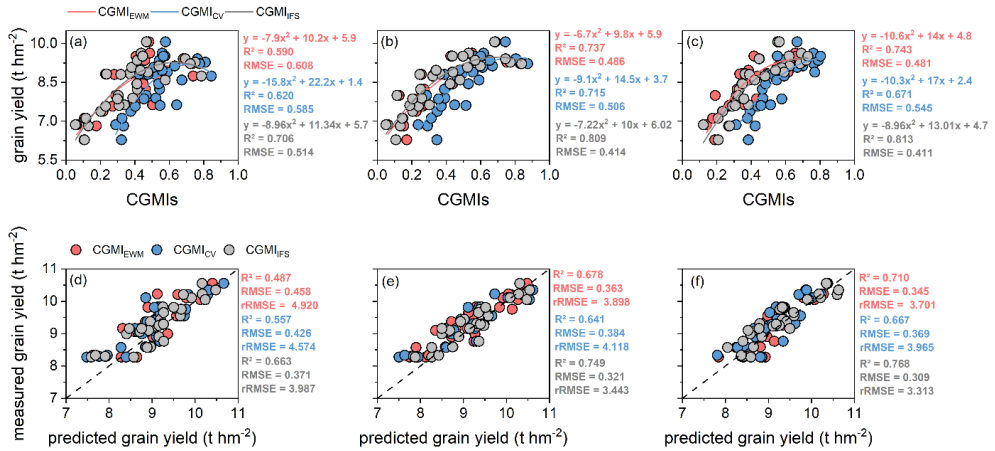


Fig. 9. Quadratic relationship between grain yield and three CGMIs at (a) and (d) tillering, (b) and (e) jointing, and (c) and (f) full-heading stages. (a)–(c) Training results of grain yield. (d)–(f) Test accuracy of grain yield.

yield at the tillering, jointing, and full-heading stages, and $CGMI_{IFS}$ had the strongest correlation, with R^2 values of 0.706, 0.809, and 0.813, respectively. The correlation between $CGMI_{IFS}$ and grain yield was enhanced with the advancement of the growth stage. The performance of the quadratic models for grain yield was tested, as shown in Fig. 9(d)–(f). The quadratic models of $CGMI_{IFS}$ at the tillering, jointing, and full-heading stages had the highest R^2 values and lowest test errors, and their rRMSE values were 23.4% and 14.7%, 13.2% and 19.6%, and 11.7% and 19.7% lower than those of $CGMI_{EWM}$ and $CGMI_{CV}$, respectively. Therefore, the IFS method can enhance the relationship between the CGMIs and grain yield, and is expected to improve the grain yield estimation.

F. Comprehensive Growth Level Classification

In this study, the comprehensive growth level classification of rice at the tillering, jointing, and full-heading stages was determined based on the quadratic fitting curves of $CGMI_{IFS}$ and grain yield using the yield-decline-ratio-based thresholding strategy, as shown in Fig. 10. Two solutions were

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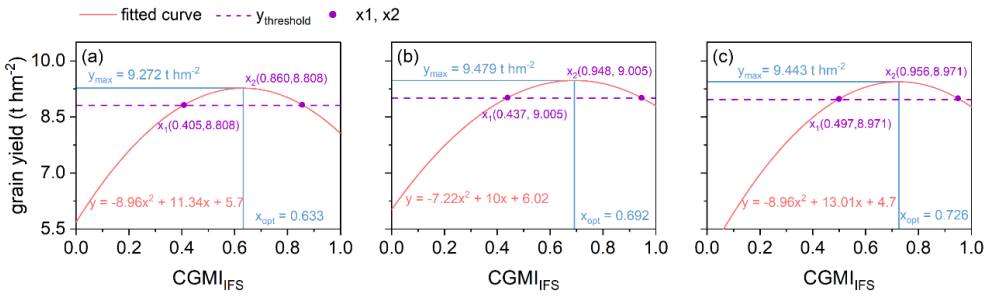


Fig. 10. Yield-decline-ratio-based thresholding strategy based on the IFS method. (a)–(c) Tillering, jointing, and full-heading stages, respectively.

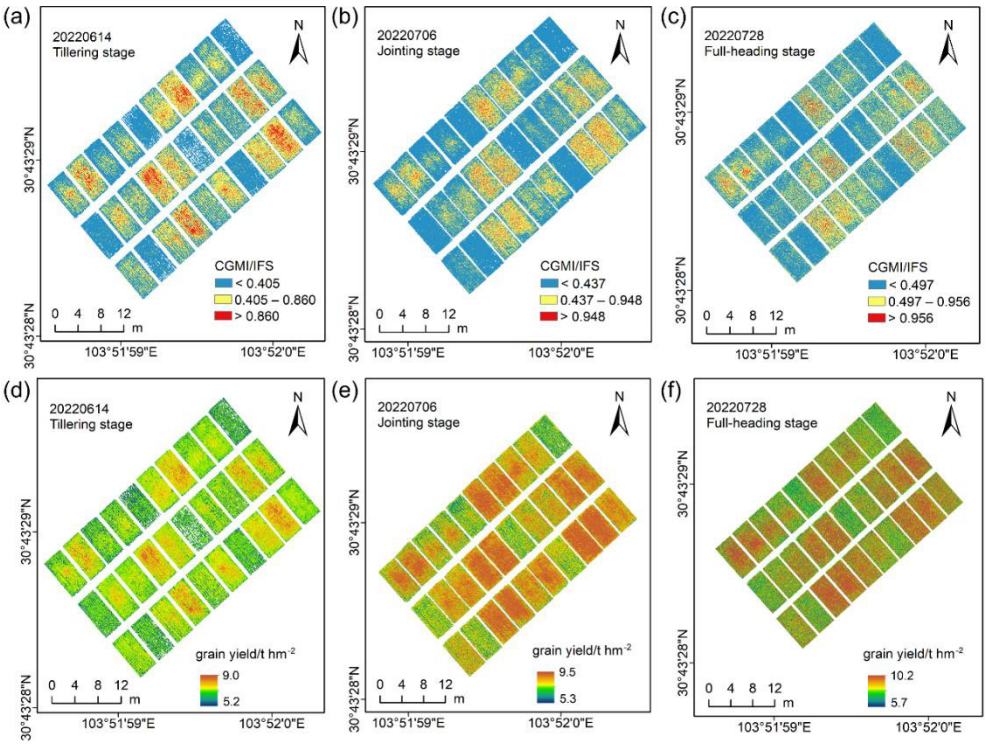


Fig. 11. Inversion of comprehensive rice growth and grain yield based on the IFS method at the tillering, jointing, and full-heading stages. In (a)–(c), blue pixels represent weak growth, yellow pixels represent suitable growth, and red pixels represent excessive growth.

calculated as the lower (minimum x_1) and upper (maximum x_2) limits of the comprehensive growth levels based on the yield thresholds on a quadratic fitting curve, and the comprehensive growth levels were classified into three categories: weak, suitable, and excessive growth (38). For the tillering stage, $CGMI_{IFS}$ values below 0.405 (weak growth: $x_1 < 0.405$) or above 0.860 (excessive growth $x_2 > 0.860$) could mean lower grain yield, whereas $CGMI_{IFS}$ values between 0.405 and 0.860 indicate that the rice growth was appropriate and that there is an expectation of achieving the target grain yield. For the jointing stage, the weak growth was $x_1 < 0.437$, the interval of suitable growth was $0.437 \leq x \leq 0.948$, and the excessive growth was $x_2 > 0.948$. For the full-heading stage, when the $CGMI_{IFS}$ value was below 0.497, rice was considered to be a weak growth; the appropriate growth was $0.497 \leq x \leq 0.956$, and the value of $CGMI_{IFS}$ exceeded 0.956, indicating that excessive growth of rice plants will reduce grain yield. As the growth stage progressed, the lower and upper limits of the comprehensive growth levels slightly rose, which was closely related to organ development and dry matter accumulation in rice.

G. Comprehensive Growth Assessment and Grain Yield Estimation Using Hyperspectral Images

UAV-based hyperspectral images were used to assess the comprehensive growth levels and to estimate the grain yield at the tillering, jointing, and full-heading stages, as shown in Fig. 10. At the tillering stage [see Fig. 11(a)], three subplots of the treatment N2M1 with excess nitrogen rate showed excessive growth, while the treatment N2M2 with standard nitrogen rate generally showed suitable growth of rice plants, and for the other nitrogen-deficient treatments (e.g., N0 and N1M1) the vast majority of the pixels assessed the rice plants as being of weaker growth. At the jointing stage [see Fig. 11(b)], 7 days after flower-promoting fertilizer application resulted

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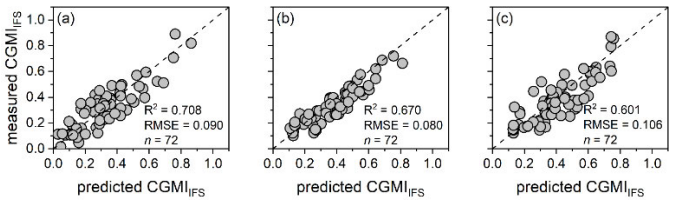


Fig. 12. Independent validation results of rice CGMI using the RF regression algorithm based on the IFS method. (a)–(c) Tillering, jointing, and full-heading stages, respectively.

in excessive rice growth of treatment N2M2, and the other rice plants maintained suitable growth. However, the treatment N2M1 gradually returned to suitable growth of rice plants due to the nitrogen fertilizer being unapplied at the jointing stage, while the other nitrogen-deficient treatments of rice plants still showed weaker growth. At the full-heading stage [see Fig. 11(c)], rice plant growth stabilized after 14 days of flower-preserving fertilizer being applied for the treatment N2M2. Rice plants of nitrogen-deficient treatments (N0, N1M1, and N1M2) were determined to grow more weakly, and rice plants of treatments N2M1 and N2M2 were more appropriately displayed.

The inversion results of the rice grain yield were checked for three growth stages based on CGMI$_{IFS}$ [see Fig. 11(d)–(f)], and found that the inversion range of grain yield at the full-heading stage was between 5.7 and 10.2 t · hm^{-2}, which is close to the actual grain yield. The inversion effects of grain yield were the weakest for the tillering stage. However, when rice plants were evaluated as a weaker comprehensive growth level for three growth stages ($x < x_1$), the corresponding grain yield was also lower. Rice plants with excessive growth at the vegetative growth stage, such as the treatment N2M1 of the tillering stage [see Fig. 11(d)], also had difficulty in achieving a reasonable grain yield. Rice plants for treatments N2M1 and N2M2 at the jointing and full-heading stages under a suitable growth level would achieve reasonable grain yield [see Fig. 11(e) and (f)]. The grain yield of some rice plants was overestimated for the nitrogen-deficient treatments (N1M1 and N1M2) at the jointing stage, and this aspect needs to be discussed further.

H. Independent Validation

The RF estimation models of CGMI$_{IFS}$ were validated using an independent dataset. The results of the independent validation are shown in Fig. 12. The model achieved R^2 values of 0.708, 0.670, and 0.601 and RMSE values of 0.090, 0.080, and 0.106 at the tillering, jointing, and full-heading stages, respectively. Compared with the test results for RF estimation models of CGMI$_{IFS}$ [see Fig. 8(b), (f), and (j)], the RMSE increased by 66.6%, 43.7%, and 58.4% across the three growth stages. This reduction in accuracy was expected because the independent dataset differed from the training dataset in terms of environmental conditions and cultivation practices. Nevertheless, the model maintained satisfactory estimation performance, indicating that the proposed IFS method possesses good robustness and transferability across different production conditions.

IV. Discussion

A. Response of Comprehensive Rice Growth for Three Methods

In this study, EWM, CV, and IFS methods were used to quantify the comprehensive growth status of rice populations, and showed significant differences in expressing growth levels, responding parameter changes, and stabilizing parameter fusion. CH and PWC had low weight coefficients in the construction of CGMI$_{CV}$ and CGMI$_{IFS}$, indicating a lack of contribution (see Table V). However, the weight coefficients of CH and PWC were not stable for CGMI$_{EWM}$; even though it had a higher weight value, it was not consistent with the change trend of CH and PWC. Therefore, the EWM method is less stable in comprehensive growth monitoring. Furthermore, the CGMI$_{EWM}$ estimation models with high test error exhibited unreliable and unstable monitoring results. As the growth process of rice plants accelerates, the CV method amplifies the effects of a few outliers due to the over-reliance on the degree of discretization of growth parameters [41], resulting in increased fluctuations of CGMI$_{CV}$, which in turn affects the ability to discriminate the comprehensive growth level. CHL and CAR provided critical information for CGMI$_{IFS}$ during the three growth stages. Photosynthetic pigments are important raw materials for photosynthesis in crop plants, which are closely related to the nitrogen status and grain yield [4], [44], and are often used to evaluate crop growth levels through hyperspectral remote sensing data [22], [23], [24], [27]. Plant leaves absorb water and carbon dioxide through photosynthesis, which promotes the accumulation of photosynthetic products and nitrogen uptake [45], an important reason why AGB, LNC, and LCC were closely related to CGMI$_{IFS}$. At the jointing stage, the weight coefficients of LNC and LCC were higher than the tillering and full-heading stages, indicating that the comprehensive growth of the jointing stage was more dependent on the coordinated expression of photosynthetic capacity and carbon and nitrogen assimilation efficiency [46]. At this time, rice is entering the transition phase of vegetation growth and reproductive growth, nutrient redistribution is intensified, and carbon and nitrogen balance have a significant impact on the development of reproductive organs [47]. The lower weight coefficients of LNC and LCC at the full-heading stage, compared with those at the jointing stage, may be related to the transfer of carbon and nitrogen substances to the grains. Therefore, the IFS method proposed in this study effectively achieves the representation of multiparameter information and optimizes the fusion contribution and physiological interpretability of growth parameters in CGMI.

B. Sensitivity of CGMIs to Hyperspectral Remote Sensing Data

The weight coefficients of the CHL, CAR, and LNC in CGMI$_{CV}$ were higher than those of CGMI$_{EWM}$ and CGMI$_{IFS}$ at the three growth stages. However, since the weight coefficients of CGMI$_{CV}$ were significantly affected by fluctuations in growth parameters, its noise was enhanced, which led to a lower correlation between CGMI$_{CV}$ and hyperspectral VIs

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than that of $CGMI_{EWM}$ and $CGMI_{IFS}$. At the tillering and jointing stages, the correlation between $CGMI_{IFS}$ and VIs was stronger than that of $CGMI_{EWM}$ and $CGMI_{CV}$. Meanwhile, the correlation between VIs and the three CGMIs decreased significantly at the full-heading stage, which might be related to the saturation effect caused by higher AGB and leaf area index [48]. Canopy structural variability increased at the full-heading stage, such as shading of rice panicles and reflectance of the leaf abaxial surface, leading to a decrease in reflectance stability [49].

Although the $CGMI_{EWM}$ estimation models had a high R^2, the test errors (RMSE and rRMSE) were larger than those of $CGMI_{CV}$ and $CGMI_{IFS}$. This phenomenon reflects that the EWM method relies on the information entropy to measure the degree of dispersion of growth parameters, which is prone to produce an assignment bias in high redundancy among the growth parameters, and thus masks the real growth status in rice. Therefore, although the EWM method can theoretically reflect the importance of the amount of information, there is an obvious disadvantage in generalization and robustness for CGMI estimation models. The IFS method showed superior accuracy across three growth stages and four regression algorithms. Therefore, the IFS method should be prioritized and recommended to construct the CGMI and to develop a robust and universal rice growth information monitoring model using hyperspectral VIs. Although the independent validation results were lower than those obtained from the test dataset, such performance degradation is common when models are transferred across locations and management conditions. Variations in environmental factors, cultivation practices, and crop growth characteristics may alter the spectral responses of VIs and consequently affect model performance. Nevertheless, the RF models of $CGMI_{IFS}$ maintained relatively stable estimation accuracy across three growth stages, suggesting that the proposed IFS-based CGMI possesses good robustness and generalizability. Future studies should investigate datasets with more locations, years, and cultivars to further enhance the generalizability and practical applicability of the IFS method.

In addition, this study focused primarily on hyperspectral VIs, which mainly capture crop biochemical and physiological information. Future research could integrate deep learning with multimodal remote sensing data, such as LiDAR and thermal infrared images, to provide complementary information on canopy structure, plant height, and crop water content. The fusion of spectral, structural, and thermal information may further improve the robustness and adaptability of CGMI estimation models, as well as enhance the accuracy of rice growth monitoring and grain yield estimation across various environmental conditions. Furthermore, the high-spatial-resolution hyperspectral images may provide additional opportunities for improving the estimation of rice growth parameters and grain yield [50]. Recent advances in deep-learning-based hyperspectral super-resolution techniques have demonstrated considerable potential for simultaneously preserving spectral fidelity and enhancing spatial detail [51], [52], [53]. Therefore, future studies should further investigate the applicability of high-spatial-resolution hyperspectral images for crop growth monitoring and grain yield prediction. Sample augmentation strategies, attention-based feature learning, and spatial–spectral–temporal modeling frameworks have demonstrated strong capabilities for extracting discriminative information from complex hyperspectral data [54], [55], [56] and will provide valuable methodological guidance for estimating CGMI and grain yield using UAV-based hyperspectral images.

C. Physiological Response of Grain Yield to CGMIs

The contribution to grain yield was explained by the $CGMI_{IFS}$, which was established by combining seven growth parameters. The effect of growth parameters on rice grain yield was not explained in this study, because their relationship has already been reported from previous studies [22], [24]. Wang et al. [22] found that higher biomass samples were calculated with lower CGMIs, which also resulted in lower grain yields. Indeed, excessive biomass is often perceived as excessive crop growth and leads to reduced grain yield [57], [58]. Therefore, a higher CGMI should be calculated when rice growth is excessive, which explains the quadratic relationship between CGMI and grain yield. The quadratic fitting curve delineated the range of $CGMI_{IFS}$ values corresponding to the relative optimal grain yield interval (see Fig. 10), and thus achieved the objective classification of the three growth levels. The yield-decline-ratio-based thresholding strategy not only enhances the agricultural interpretability of CGMIs but also provides a quantitative reference for nitrogen regulation in the rice field. However, from the distribution of the evaluated boundaries, the lower (x_1) and upper (x_2) limits of $CGMI_{IFS}$ showed an elevated trend with growth stage, increasing from [0.405, 0.860] at the tillering stage to [0.497, 0.956] at the full-heading stage, which reflected the growth and developmental behavior inherent to the different growth stages and developmental trends toward ideal population dynamics. The elevation of the lower threshold of $CGMI_{IFS}$ explains the minimum biomass and physiological activity necessary for the plant to reach that growth stage in order to satisfy the minimum requirements for organogenesis and grain formation and to avoid irreversible losses due to growth lag. In terms of plant physiology, the increase in the upper threshold indicates the requirement for stronger growth and higher biomass during the reproductive growth phase of rice to ensure the photosynthetic capacity, organ development, and material accumulation required for high grain yield, while preventing the risk of grain yield reduction due to overgrowth [59]. The shortcoming of the yield-decline-ratio-based thresholding strategy is that it does not consider the dynamic adjustment effects by regions, meteorology, and cultivars, which may lead to poor adaptation between years. The comprehensive growth level classification strategy based on $CGMI_{IFS}$ and grain yield provides an actionable regulation target for field nitrogen management, but it is still necessary to combine multisource data, process models, and multidimensional parameters to construct a more dynamic, precise, and intelligent growth classification framework to promote the practical production application in rice.

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V. Conclusion

In this study, an IFS method based on cluster stability and a pseudo-supervised fusion mechanism was proposed to construct a rice $CGMI_{IFS}$ by combining seven growth parameters and compared with EWM and CV methods. The effectiveness of $CGMI_{EWM}$, $CGMI_{CV}$, and $CGMI_{IFS}$ was assessed in monitoring the comprehensive growth status of rice using UAV-based hyperspectral images. A yield-decline-ratio-based thresholding strategy was used to classify rice growth levels and to explore $CGMI_{IFS}$ response to nitrogen stress and grain yield estimation. The main conclusions of the study are summarized as follows.

1) The relationships between seven growth parameters and the three CGMIs were analyzed at the tillering, jointing, and full-heading stages, and $CGMI_{IFS}$ had stronger discriminatory ability and stability than $CGMI_{EWM}$ and $CGMI_{CV}$, and especially performed more prominently in reflecting changes in key physiological indicators, such as chlorophyll and nitrogen contents.
2) Seven VIs were selected by the permutation importance algorithm to develop the estimation models of rice CGMIs. The RF regression algorithm showed the best estimation performance for the three CGMIs, where $CGMI_{IFS}$ had stronger test accuracy than $CGMI_{EWM}$ and $CGMI_{CV}$ (tillering: R^2 = 0.895, RMSE = 0.030, and rRMSE = 7.275%; jointing: R^2 = 0.901, RMSE = 0.045, and rRMSE = 11.364%; full-heading: R^2 = 0.788, RMSE = 0.044, and rRMSE = 9.318%).
3) Three CGMIs showed a quadratic relationship with grain yield at three growth stages, with $CGMI_{IFS}$ having the strongest estimation ability of grain yield. The yield-decline-ratio-based thresholding strategy was used to solve two critical solutions (x_1, x_2) from the quadratic fitting curve and to determine the comprehensive growth levels, achieving a quantitative discrimination of comprehensive growth status. At the tillering and jointing stages, excessive or weak rice growth may reduce grain yield, whereas appropriate $CGMI_{IFS}$ values correspond to higher grain yield.
4) Spatial inversion of rice $CGMI_{IFS}$ was performed using UAV-based hyperspectral images. $CGMI_{IFS}$ could effectively recognize the changes in comprehensive growth status due to nitrogen rates, and the inverted grain yields reached the highest accuracy at the full-heading stage, which was highly consistent with the real grain yield. Particularly, the IFS method demonstrated good responsiveness for identifying growth anomalies and grain yield differences caused by nitrogen deficiency and over-fertilization.

Therefore, the CGMI constructed using the IFS method showed higher fusionality, physiological interpretability, and application potential under multiple growth stages and provided a generalizable, quantifiable, and operational comprehensive evaluation framework for rice comprehensive growth monitoring.

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